

Forecasting British elections: a dynamic perspective

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Abstract

This is a plea for a dynamic perspective in forecasting British elections. Autoregressive models are capable of making forecasts in their own right (ex-ante, early, and unconditionally). Their large forecast errors, however, suggest that these models be used in combination with structural models of the vote. Lewis-Beck has identified the key short-term predictors of the vote such as government approval and economic conditions. The performance of such a vote model can only be helped by the inclusion of a dynamic element that captures the undeniable return of the British vote to equilibrium.

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Election forecasting is too important to leave to pollsters. Theories of electoral behavior have gone a long way toward producing parsimonious models that have been put to the test with surveys of individual voters or aggregate electoral data. Lewis-Beck stands out as a pioneer in efforts to compress electoral insights into forecasting models. Having shown the way with his predictions of US presidential contests as well as French elections, he has now trained his sight on Britain. His model for British elections proves highly successful, picking the winner ten out of twelve times. And it does so with only three predictors, key among them being voter satisfaction with incumbent government performance.

Yet structural vote models of this kind are not the only alternative to poll-based forecasts. However peculiar each election may appear to be, these are regular events

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occurring in a sequence that is not likely to obey the rules of a coin toss. Are there cycles to winning and losing elections? Does the partisan vote exhibit features of equilibrium? A dynamic model of the vote incorporating these elements may provide important clues about future election outcomes. What is more, such a model is capable of coming up with true *ex-ante* forecasts, in the sense that they are available long before the event. They do not require any information about conditions known with certainty only after Election Day. But like everything, this advantage has its price, too.

1. Parameters of the major-party vote in Britain

Britain has a history of general elections that is long enough to permit the application of dynamic models. For the most part, the electoral rules of the game and the identities of the major political parties have remained the same for almost two centuries. The size of the electorate, of course, has vastly expanded since 1832, and one of the leading parties in the 19th century sharply declined in electoral standing during the 20th century. At the same time, a party not present in the 19th century rose to prominence in the 20th century. Party competition in the British parliament nonetheless has rarely deviated from a two-party pattern. Under an electoral system of plurality voting that is what one should expect to happen (Duverger, 1963; Rae, 1971). From 1832–1918, the party battle of Britain was one between Conservatives and Liberals, followed by one between Conservatives and Labor.¹ The rise of the new party precipitated a quick demise of one of the older parties, with the Conservatives the lucky ones to survive this upheaval without damage.

Figure 1 charts the Conservative percentage of the major-party vote, with the Liberals being the other major party until 1918, and Labor being that party afterwards (as marked by the vertical line in Fig. 1). Labor convincingly overtook the Liberals in the election of 1924 and has kept them in third place ever since. Hence, with that election the two-party set switches from Conservative-Liberal to Conservative-Labor. To chart the Conservative percentage of the major-party vote over such a long stretch of time, still another adjustment was required. In general elections during the 19th century, it was quite common for many constituency races to be uncontested. With no votes reported for those constituencies, the national party total, based only on contested races, is distorted. That matters greatly in elections where one of the parties had a decided edge in uncontested seats. With that cushion it could afford to lose the popular vote while still managing to win a majority of seats in parliament. For elections where exactly that happened (1847, 1852, and 1874), this analysis has applied a correction to the popular vote. The correction relies on an estimate of the

¹ The data for the party vote in Britain come from Craig (1981, p. 49), King (1993, p. 249), Hoge (1997), and Clarke et al. (2001). In those instances where two elections were held in quick succession, only the second one was used in most cases, leading to the exclusion of the elections of 1835, 1857, 1885, 1950, 1964, and Feb. 1974; in the more complicated instance where three elections took place in successive years, the middle one (1923) was dropped.

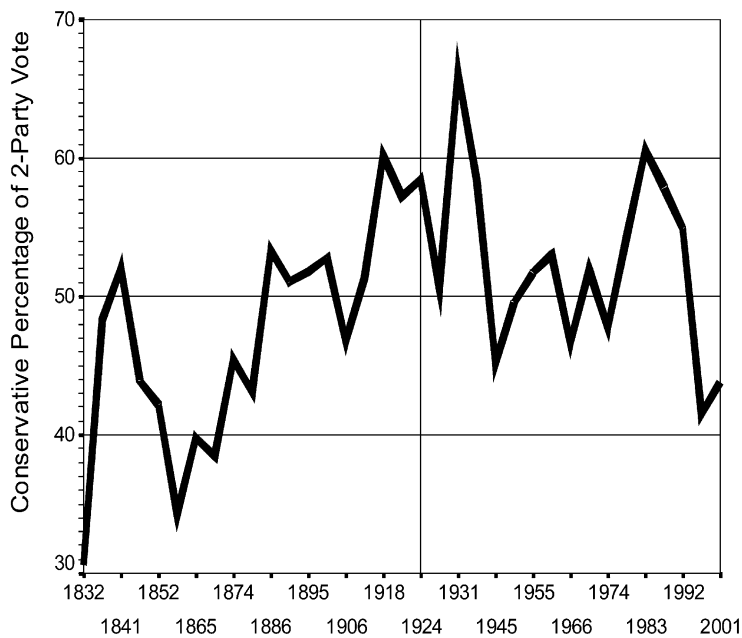


Fig. 1. The Conservative vote in British general elections, 1832–2001

vote-seat relationship using all elections except the three suspect ones.² Based on the estimates for that relationship, derived values were entered for the Conservative vote for the elections of 1847, 1852, and 1874.

As charted in Fig. 1, a vote share above the 50% mark indicates a Conservative victory, in terms of the national vote, and a share below that indicates electoral defeat. What attributes does the flow of partisan division in British general elections over nearly two centuries exhibit? There are three features worth noting. One is the bounded nature of the vote. In winning, the Conservatives never topped 67% of the two-party vote, nor did they fall below 31% in losing. And these are quite extreme markers. For the most part, the vote stays within a 20-point zone between 40 and 60%. The second feature worth noting is the average of the Conservative vote over the 1832–2001 period: at 49.6%, it is practically indistinguishable from the 50% mark. Given the focus on the two-party vote, that holds true, of course, for the other major party as well. This equality is not the result of offsetting inequalities, where one party is dominant for a long stretch followed by an equally long stretch of electoral dominance by the other party. Instead, as indicated by the third feature of the flow of the vote in Fig. 1, electoral turnover is quite frequent in British general elections. With the vote crossing the 50% line twelve out of 35 times during the

² The equation specifies a linear relationship between the Conservative vote percentage (V) to the Conservative seat percentage (S). The estimates are: $V = 28.7 + 0.44S$. $R^2 = 0.71$; $N = 31$. This equation leads to the following estimates of the Conservative vote: 49.6 (1847), 48.7 (1952), and 52.5 (1874).

1832–2001 time frame, a major party in Britain has typically won no more than three general elections in a row before being ousted at the polls. Neither the Conservatives nor their major-party opponent have enjoyed prolonged ascendancy in British elections. These features add up to a strong *prima facie* case that the major-party vote division in Britain is in a state of equilibrium.

2. Electoral equilibrium in Britain

The defining characteristic of equilibrium is stability. Equilibrium means stable solution. In game theory, where this concept occupies a central role, “behavior at an equilibrium is stable in the sense that no actor, given its current position and knowledge, can improve its own position on its own.” (Morrow 1994, 8) One of the first to introduce the notion of equilibrium in the study of elections was Downs (1957). In his theory of party competition, equilibrium is a function of both voter preferences and party strategies, with electoral systems getting some credit as well.

If the distribution of ideologies in a society’s citizenry remains constant, its political system will move toward a position of *equilibrium* (emphasis supplied) in which the number of parties and their ideological positions are stable over time. (Downs 1957, 115)

For an electorate with a unimodal distribution of ideological positions in a one-dimensional policy space, Downs predicted that equilibrium would be reached by two parties pursuing a strategy of ideological convergence, while operating under plurality-voting rules. On the other hand, in a society with several modes of ideological positions and proportional-representation rules, a multiparty system would produce equilibrium where the various parties — one for each ideological mode — would strive for maximum ideological differentiation. Unlike the multi-party, equilibrium in the two-party case implies not only long-term stability, but also equal balance between the two parties. That is clearly the sense in which Stokes and Iversen (1966), in their analysis of US party competition, apply the concept of equilibrium:

We use the term ... ‘equilibrium’ force as a generalized name for factors tending to return the party division to 50% [of the two-party vote] in the long run. (Stokes and Iversen 1966, 184)

It is an altogether different world where equilibrium is missing. In that universe, as Heraklitus put it, everything is in flux. An important model capturing movement without equilibrium is the random walk. Stokes and Iversen (1966) introduced this model as a means of testing for electoral equilibrium. Rejection of the random-walk model proved the existence of equilibrium, that is, “the existence of forces restoring party competition.” In a random walk, there is no “restoration” of an underlying balance. Instead, from one election to the next one the vote (V_t) is able to move in random fashion (u_t), like a particle in a space without gravity.

$$V_t = V_{t-1} + u_t \quad (1)$$

Such behavior may not reveal instability instantly, although the reference to it as “drunkard’s walk” conveys its menace. While a drunk is able to take only small steps, he may end up far away from where he started. There is no correction of steps taken, nothing to steady the movement. In other words, there is no return to gravity. As for a party that fared badly in one election, its prospects of doing better the next time are no better than the prospects of dropping even further. Worst, there is no assurance of survival in the long run. A standard test for a random walk is provided by the Dickey-Fuller procedure (Dickey et al., 1986).³ The rejection of that hypothesis confirms the existence of equilibrium. Failure to reject the hypothesis implies lack of equilibrium. The test result for the major-party vote in Britain ($t = -3.819$) allows us to reject, without any hesitation, the random walk model. The chances are less than one in 100 that the major-party vote in British general elections all the way back to 1832 could have moved the way it did in the absence of equilibrium.

It may seem paradoxical that a predictable consequence of electoral equilibrium is partisan turnover. Just like the French saying, *plus ça change, plus c’est la même chose*. Under equilibrium electoral change is frequent and hence something to expect and to predict. A party taking control is bound to lose it in subsequent elections. The only question is how quickly the loss occurs. That is a lot less predictable unless winning and losing elections is a cyclical phenomenon.

3. A dynamic forecast of the British vote

The tendency of a government to lose electoral support in subsequent elections has drawn much attention in studies of voting (Campbell and Converse, 1960, ch. 20). Seen in light of economic theory, some have construed that loss as a “cost of ruling” (Paldam, 1991). It must be noted that Lewis-Beck includes a measure of the cost of ruling in his forecasting model of British elections. The measure takes the form of a linear decline. That way of handling the return of the vote division to equilibrium, however, may prove too inflexible for optimal forecasting. The alternative is to view the return to equilibrium as a dynamic process following the path of an autoregressive, moving-average (ARMA) model (Box and Jenkins, 1976; Clarke et al., 1998).

The pattern formed by the autocorrelations and partial autocorrelations for the

³ The specific form of the test is applied to a random walk model written as $\nabla V_t = \lambda V_{t-1} + u_t$. This equation relates the change of the vote between two elections to the vote in the previous election. The logic of the random walk implies a null relationship, hence $\lambda = 0$. In a random walk, a high vote in one election does not trigger a decline, nor does a low vote trigger an increase in the next election. Equilibrium, on the other hand, implies that such adjustments occur in the direction of the mean level of the process.

major-party vote in Britain (Fig. 2) strongly points to a simple autoregressive dynamic. There are no hints of a more complex process, or of any cyclical patterns. In other words, the winning party will lose its edge in subsequent elections, but there is no way to predict *when* exactly, that is, after how many terms the winner will lose the majority (of the major-party vote). The estimates for the AR(1) model of the Conservative vote (V_t) in Britain from 1832–2001 are:

$$V_t = (1-0.57) 48.7 + 0.57 V_{t-1} + u_t \quad (2)$$

(2.5) (0.14)

$$\text{SER} = 6.6 \quad \text{Adj. } R^2 = 0.25 \quad \chi^2_{(8)} = 3.5 \quad N = 35$$

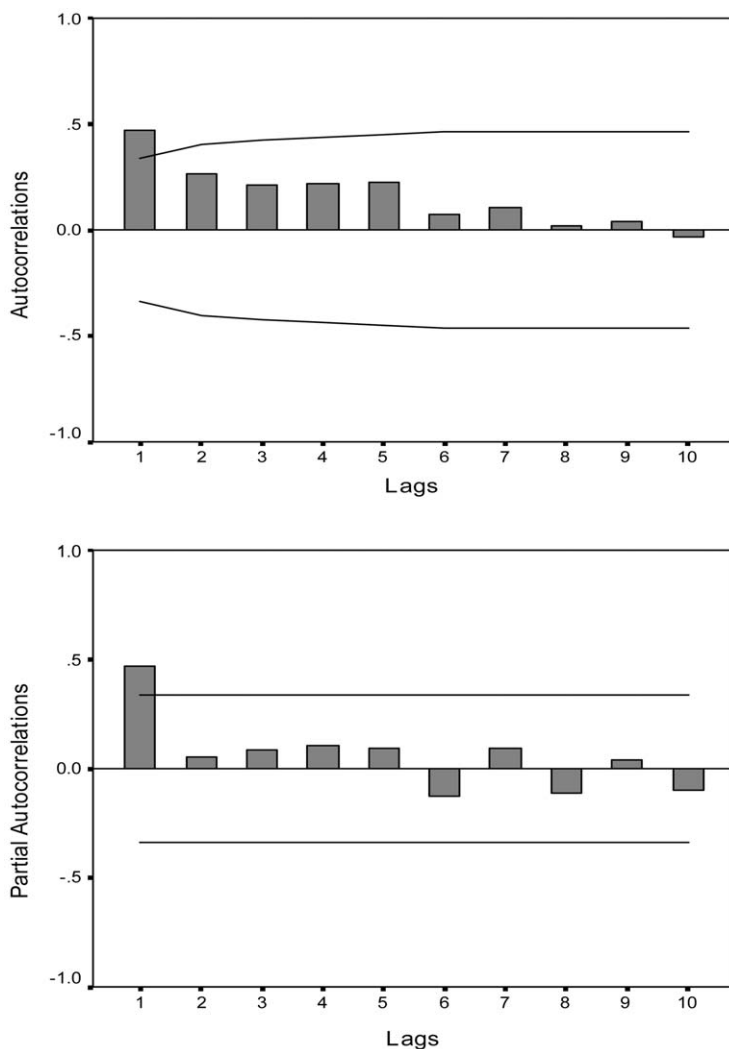


Fig. 2. Autocorrelations and partial autocorrelations for the Conservative-vote series.

With an autoregressive parameter of 0.57, one can predict a quick return to equilibrium in British elections. The winning party should expect to retain only about half of its above-equilibrium vote share at the next election, and about half of the remainder at the election after that one, and so on. The winning party's half-life, in other words, barely exceeds one term.

Figure 3 charts the model predictions along with the actual Conservative vote for British general elections from 1832–2001. To be sure, these are in-sample predictions, based on the model estimates for the full period, not out-of-sample forecasts, based solely on prior observations. The comparison of the actual vote (light bars) and the predicted vote (dark bars) is instructive nevertheless. One can see that the predictions are sharply off in years when control changes between the major parties. For 1997, for example, the model predicts 52.2% for the Conservatives, who wound up losing the election with a share of 41.5%.

For the next British general election, to be held in 2006 at the latest, the AR(1) model makes the following forecast, as far as the major-vote division is concerned:

Conservatives: 45.9%

Labor: 54.1%

In plain English, Labor will win again, though less comfortably than in 2001, when it took 56.2% of the major-party vote.

Dynamic forecasts of this sort have both a big advantage and a big disadvantage

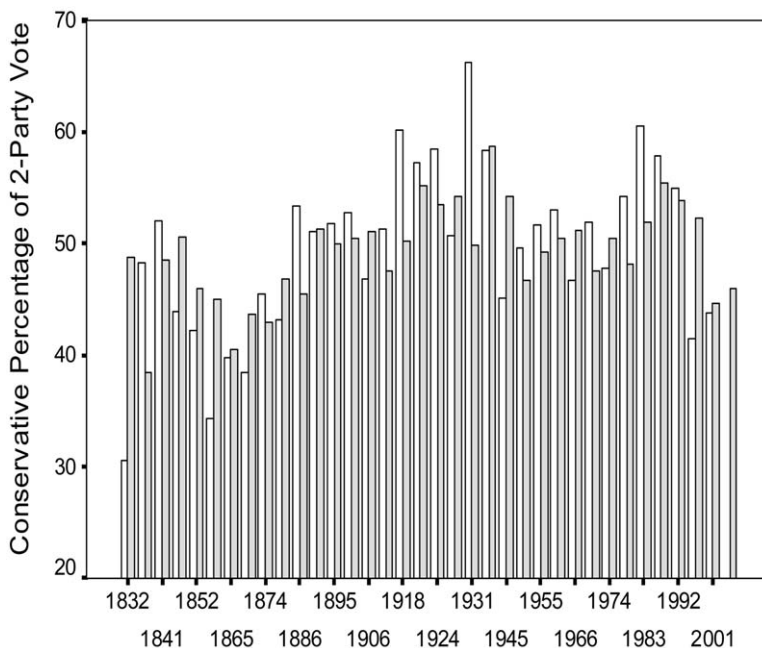


Fig. 3. Actual and predicted vote for Conservative party, 1832–2001.

over forecasts derived from structural models. The advantage, as can be seen from the exercise above, is the ability to deliver a forecast of an election outcome many years in advance. In 2001, the dynamic model could make a forecast about 2006, and without any ifs and buts. No structural vote models can do so unconditionally, that is, without assuming what the key predictors (economic indicators, government approval etc.) will be at election time. ARIMA models have no such problem. All the data needed for the forecast of a future election are known with certainty right after the last election.

This advantage, however, comes with a hefty price tag. And that is the forecast error. There is no hiding the fact that the AR(1) model of the British vote has a standard error of 6.6 percentage points. The resulting 95% confidence interval stretches beyond ± 10 points. That will be too wide for comfort for too many election forecasters. It might be wise to attach the following warning to these forecasts: do not bet your fortune on them.

The less risky way of making use of dynamic models for forecasting might be to combine them with structural models (Norpoth, 2000). To be sure, that approach surrenders the big advantage of dynamic models but it would make the margin of forecast error more palatable. The structural vote model proposed by Lewis-Beck can be easily amended to incorporate a dynamic component. In fact, that component would replace the “governing terms” variable of the model.

4. Conclusion

This paper has made a plea for a dynamic perspective in forecasting British elections. While dynamic models can make forecasts in their own right (ex-ante, early, and unconditionally), their large forecast errors make them risky ventures. More acceptable may be their use in combination with structural model. Lewis-Beck has identified the key short-term predictors of the vote such as government approval and economic conditions. The performance of such model can only be helped by the inclusion of a dynamic element that captures the undeniable return of the British vote to equilibrium.

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